Course Code : ECT 258 **MQNR/RS – 23 / 1182**

Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

MICROPROCESSORS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated.
(2) Solve any **Three** questions from each part.
(3) Due credit will be given to neatness and diagrams.
(4) Use of electronic gadgets leaving scientific calculator is not allowed.
(5) Assume suitable data wherever necessary.

PART A

1. (a) Recall Pins of 8085 microprocessor and explain the significance of following pins in brief :

(a) $\overline{\text{RESET IN}}$
(b) ALE
(c) IO / $\overline{\text{M}}$
(d) Trap
(e) X1, X2 5(CO1)

(b) Illustrate hardware interrupts from the below list, also infer how is the address of the Interrupt Service Routine calculated for the vectored interrupts with calculations.

List if Interrupts : RST 0, RST 2, RST 3, RST 4·5 and RST 6·5. 5(CO3)

2. (a) Using an assembly language program that transfers 200 bytes of data from EEPROM to RAM using indirect addressing mode inside loop. Also calculate the total number of T states your program will require.

(Assume source address as 2000H and destination address as A000H respectively) 5(CO2)

(b) Integrate with an interfacing diagram for 8085 microprocessor for interfacing memory such that it should satisfy the **above question 2 (a)**, EEPROM and RAM configuration using absolute decoding technique. 5(CO2,3)

3. (a) Deconstruct indirect addressing mode instruction STAX B with neat Timing diagram / Waveform and access the benefits of the said instruction over direct addressing mode in brief.

   (Assume BC pair has contents ABCDh and program counter is at 1234h). 5(CO1,3)

   (b) Judge the interfacing requirements of 8085 microprocessor and 8255 PPI to interface a seven segment display at port A of 8255 by deconstructing with a diagram and a program to display "9" continuously. **NOTE : All calculations are to be shown.** 5(CO3,4)

4. Design the system of 8085 microprocessor in a way to interface a stepper motor with 8255 along with consideration to rotate stepper motor 30 degree clockwise at 50 RPM and 45 degree counter clockwise at 100 RPM continuously, write a program. **NOTE : All calculations are to be shown**. 10(CO2,4,5)

PART B

5. (a) Retrieve with a neat diagram and explain the flag structure of 8086. 5(CO1)

   (b) Attribute what you understand by real and virtual memory in detail. 5(CO4)

6. (a) Recall and explain the important features of Intel 286 microprocessor. 5(CO4)

   (b) Illustrate the working of following 8086 instructions :

   (i) MOV BX, AX

   (ii) ADD AX, [DI] 5(CO2)

7. (a) Using example instructions explain the working and need of stack memory in 8086 microprocessor. 5(CO2,3)

(b) Deconstruct 8086 Assembly Language program to multiply two immediate bytes 45h and 69h and then count the number of 0's (zero bits) present in the result. 5(CO2)

8. Critiquing the algorithm to arrange 10 bytes of data in memory 5000h onwards by judging codes for both 8085 as well as 8086 microprocessors in terms of code optimization. 10(CO2,4,5)